

Indirect Measurement

Finding a pattern is a useful problem-solving strategy. It can save time you might otherwise spend in lengthy calculations. In this lesson, you will see how patterns can be helpful in solving problems.

One example of patterns in mathematics is the **number series**. A number series is a set of numbers that continues according to a rule.

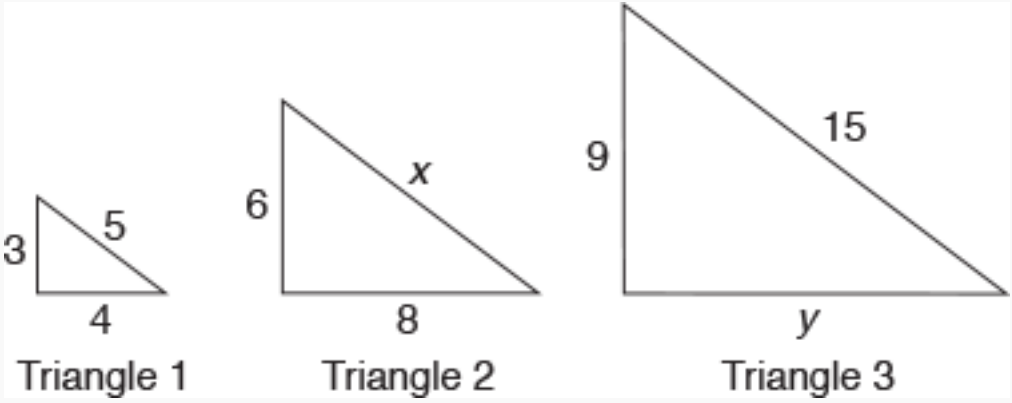
The counting numbers (1, 2, 3,...) are a number series in which each number is 1 larger than the number before it. In the series 1, 4, 7, 10,..., each number is 3 more than the preceding number.

You have seen how you can use proportion to solve problems that involve similar figures. Finding a pattern can often get the job done just as well. In the next example, the information is organized in a table. Tables and charts can make patterns easier to recognize.

Example

Hide

The triangles are similar. Find the measures of the missing sides.



	Vertical Leg	Horizontal Leg	Hypotenuse
Triangle 1	3	4	5
Triangle 2	6	8	x
Triangle 3	9	y	15

Read down each column to find the pattern. You can quickly see that the sides of Triangles 2 and 3 are multiples of Triangle 1.

The missing side in Triangle 2 is **10**. The missing side in Triangle 3 is **12**.

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Lesson 6: Problem Solving with Triangles

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